MIL-C-4551E
3 February 1971
SUPERSEDING
MIL-C-4551D
30 September 1964

MILITARY SPECIFICATION

CRANES, FLOOR, PORTABLE

This specification is mandatory for use by all
Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers portable floor cranes.

1.2 Classification. The cranes shall be of the following groups, types, classes and styles (see 6.2):

GROUP A - 4,000 Pound Capacity

GROUP B - 3,000 Pound Capacity

GROUP C - 2,000 Pound Capacity

Type I - Mechanical Lift
Type II - Hydraulic Lift

Class 1 - Rigid Legs
Class 2 - Adjustable Legs

Style (a) - Extensible Boom
Style (b) - Non-Extensible Boom

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-T-81	Tags, Shipping and Stock
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication

FSC 3950

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods

(Copies of specifications, standards, drawings and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 *Screw-Thread Standards for Federal Services*

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Il 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc. Attn: Tariff Order Section, 1616 P Street, N. W., Washington, D.C. 20036.)

3. REQUIREMENTS

★ 3.1 Preproduction. This specification provides for preproduction testing.

3.2 Materials. Materials used in the construction of the cranes shall be as specified herein. Materials not definitely specified herein shall be sound, of uniform quality and condition and free of seams, cracks, and other defects which may adversely affect the appearance, strength, endurance, or wear resistance of the finished product. Materials shall not be hammered, filed, or otherwise treated to conceal defects therein. Materials shall conform in composition, heat treatment, and suitability to the standard practices of reputable manufacturers producing equipment of this type.

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* 3.3 Design. All parts and components of the crane shall be new, of first class commercial quality, and of the latest commercially accepted design. All parts shall be of such size, material, and strength necessary to sustain a 25 percent minimum overload (static load). The cranes shall operate satisfactorily in a temperature range from 0 to 125 degrees F. The base of the crane shall be low and shall be open at the end opposite from that on which the frame or boom is mounted to permit positioning the crane directly over objects.

3.3.1 Equipment and component parts. The equipment and component parts, except as specified herein, shall be essentially the standard product of the manufacturer so that prompt and continuing service and delivery of repair or maintenance parts may be assured. The component parts of the unit need not be the product of the same manufacturer.

* 3.3.2 Lubricants. Unless otherwise specified (see 6.2) the equipment shall be designed to operate efficiently and satisfactorily when lubricated in accordance with MIL-STD-838.

3.4 Construction. The crane shall be equipped with a boom, a mast, legs, wheels, casters, a floor brake, chain or wire rope, a hook, and a towing handle. The crane shall be designed for applications that primarily require straddle-lifting, such as the lifting and removing of engines from various types of gasoline and diesel-powered automotive vehicles.

* 3.4.1 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with means for lubrication. Wheels and swivel casters shall be equipped with fittings in accordance with MIL-F-3541 for efficient lubrication. Prior to delivery to the Government, all moving parts of the unit requiring lubrication shall be adequately lubricated (see 3.3.2) with lubricants of the grade and temperature range specified in the contract or order (see 6.2). After lubrication, each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.4.1.1 Caution plates. Where the use of high-pressure lubricating equipment, 1,000 pounds per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.4.2 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders, metallic pastes, or non-metallic pastes be used on castings or forgings for reclaiming any parts of the bare.

3.4.3 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.4.4 Fastening devices. All screws, pins, bolts and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.4.5 Surfaces. All surfaces of all castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

* 3.4.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.5 Components.

* 3.5.1 Handle. The towing handle shall provide an adequate means for both steering and moving the crane. The pull bar shall be located 32 to 40 inches above the floor. The handle shall fold to the base or the mast of the crane when not in use and shall be secured thereto by means of a clip or other suitable device.

* 3.5.2 Base and frame. Base and frame shall be of fabricated steel. The assembled unit shall be capable of sustaining a 25 percent (static) overload without permanent deformation. Dimensions (in inches) shall be as follows:

	Dimension Identification	Group Designation A	B	C
	Horizontal clearance between centerline of lift and any part of crane (with boom level and, if style a, retracted to maximum capacity lift position) (min)	39	39	35
	Roll-under clearance (max)	11	11	9
	Roll-between clearance (Overall width of legs) (max)	40	40	38
	"Straddle" at centerline of lift for class 1 and class 2 with legs closed and boom level (min)	22	22	22
	"Straddle" at centerline of lift for class 2 with legs spread and boom level (min)	47	47	45
**	High position of boom end (boom, if style a, retracted to maximum capacity lift position) (min)	108	108	106
**	Low position of boom end (boom, if style a, retracted to maximum capacity lift position) (min)	44	44	40

** NOTE: The point of reference is the eye of the boom tip (not considering any chain).

* 3.5.3 Floor brake. A floor brake shall be provided. A pedal or lever for safely and conveniently engaging and disengaging the brake shall be located at the mast end of the crane. Brake shall lock when engaged and require positive action for disengagement.

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* 3.5.4 Casters and wheels. Casters and wheels shall have metal tread and shall be of the type (rigid and swivel) to provide ease of maneuverability and tracking when crane is operating under full load with outriggers located in any of their various positions or fixed position, whichever is applicable. The strength, quantity, and size of casters shall be such that under all conditions of loading, up to and including rated load of crane, caster loading shall not exceed published load rating of caster manufacturer(s). In no case shall the caster wheel diameter be less than six inches for Group A, five inches for Group B, and four inches for Gourp C cranes.

* 3.5.5 Chain or cable and hook. Cranes shall be equipped with either an alloy or steel chain or wire rope, as specified (see 6.2). The chain or wire rope shall be equipped with a steel swivel type safety hook. The chain or wire rope and the hook shall have a safe working load rating at least equal to the specified crane capacity (see 1.2) in accordance with generally accepted industry design standards. When the boom is in the low position the chain or wire rope shall be of sufficient length to reach from the boom to the floor but in no case less than 18 inches long. The crane shall be equipped with the attachment on the boom which will permit adjustment of the suspended length of chain or wire rope. The swivel and hook shall not reduce the headroom more than six inches from the dimensions in high and low positions of 3.5.2.

3.6 Structure.

* 3.6.1 Type I crane. Type I cranes shall consist essentially of a steel frame with a cantilever boom at the top, mounted on a wheel supported base. A manually operated hoisting device shall be included. All parts necessary for satisfactory operation shall be provided.

3.6.1.1 Hoisting mechanism. The hoisting mechanism for the Type I crane shall have a multiple chain and block system with a crank drive or endless chain drive, or a worm and wheel with cable. Antifriction bearings shall be used throughout the crane. The mechanism shall have an overall mechanical advantage of not less than 200. The hoisting mechanism shall be equipped with safety devices to prevent slipping of the load and to require positive effort to lower the load. Self-locking worm gears shall not require additional protective devices.

* 3.6.2 Type II crane. The Type II crane shall consist essentially of a steel column with one end of a lifting boom pinned to the tip, mounted on a wheel-supported base. The hoisting beam shall be operated by a hydraulic piston mounted on the column or base. The hydraulic pressure shall be manually produced. All parts necessary for satisfactory operation shall be provided. It shall be capable of handling the rated load with safety and dispatch.

3.6.2.1 Hydraulic system. The hydraulic system for the Type II crane shall be capable of withstanding the hydraulic pressures necessary to lift loads of rated capacity, without damage or leakage. The crane shall be furnished with the hydraulic system filled with hydraulic oil of an operational type recommended by the manufacturer. The system reservoir shall hold sufficient oil to lift the rated load to its greatest height with adequate reserve capacity.

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3.6.2.2 Hydraulic hand pump. The hydraulic hand pump shall be of the double action type and shall be manually powered by means of hand lever furnished with the crane. It shall be designed and constructed so that the force on the hand lever can be quickly and easily reduced to lift heavy loads, and so that the speed of lifting light loads can be quickly and easily increased. Maximum force on the hand lever required to lift rated load at maximum speed shall not exceed 60 pounds for the 2-speed type of pump nor 80 pounds for the variable speed type of pump.

3.6.2.3 Cylinder. The cylinder walls shall be honed to a fine finish and shall be of fine-finished steel tubing. Packing shall be sufficient to prevent oil leakage. The piston shall be cold rolled steel.

* 3.6.2.4 Hydraulic system controls. Hydraulic system controls shall include a low pressure relief valve which shall relieve at 100 percent to 105 percent of rated load pressure, a check valve, a flow control valve to prevent a sudden dropping of the load in the event of system failure, a high pressure relief valve which shall relieve at 120 to 125 percent of rated pressure, and a manual lowering valve. The high pressure relief valve and the lowering valve may be combined in a single control unit. The purpose of the low pressure relief valve is to prevent raising greater than rated loads. The purpose of the high pressure relief valve is to relieve shock loads on the hydraulic system. A typical schematic of the controls system arrangement is shown in Figure 1. Pressure relief valves shall be rendered inoperative during the tests conducted in accordance with paragraph 4.4.4. All valves shall be constructed so that no sticking or freezing is possible. All hydraulic system controls shall be capable of passing the preproduction inspections and tests set forth in Section 4.

* 3.6.3 Class 1 cranes. Class 1 cranes shall have either adjustable or non-adjustable outrigger legs. If adjustable, they shall have an indexing feature to maintain the "roll between" and "straddle with legs closed" dimensions of 3.5.2.

* 3.6.4 Class 2 cranes. Class 2 cranes shall have adjustable outrigger legs. They shall be capable of being adjusted to and locked in the "roll between/straddle with legs closed" dimensions of 3.5.2. They shall also be capable of being adjusted to and locked in the "straddle with legs spread" position of 3.5.2 and at least one intermediate position.

* 3.6.5 Style (a) cranes. Style (a) cranes shall be equipped with a boom of the extensible type and shall be provided with a positive means of holding the boom in at least three positions beyond the "horizontal clearance" dimension in 3.5.2. Booms shall be capable of quick and easy adjustment without the use of tools. The extensible part of the boom shall remain attached to the main boom at all times. Groups A and B extensions shall add at least 15 inches and Group C shall add at least 12 inches to the "horizontal clearance" dimension of 3.5.2.

* 3.6.6 Style (b) cranes. Style (b) cranes shall be equipped with either an extensible or a non-extensible boom. If extensible it shall have an indexing feature to maintain the "horizontal clearance" dimension of the high and low positions of 3.5.2.

* 3.7 Performance. The crane shall have the specified working capacity (see 1.2) and shall be capable of withstanding a total suspended load (static) of

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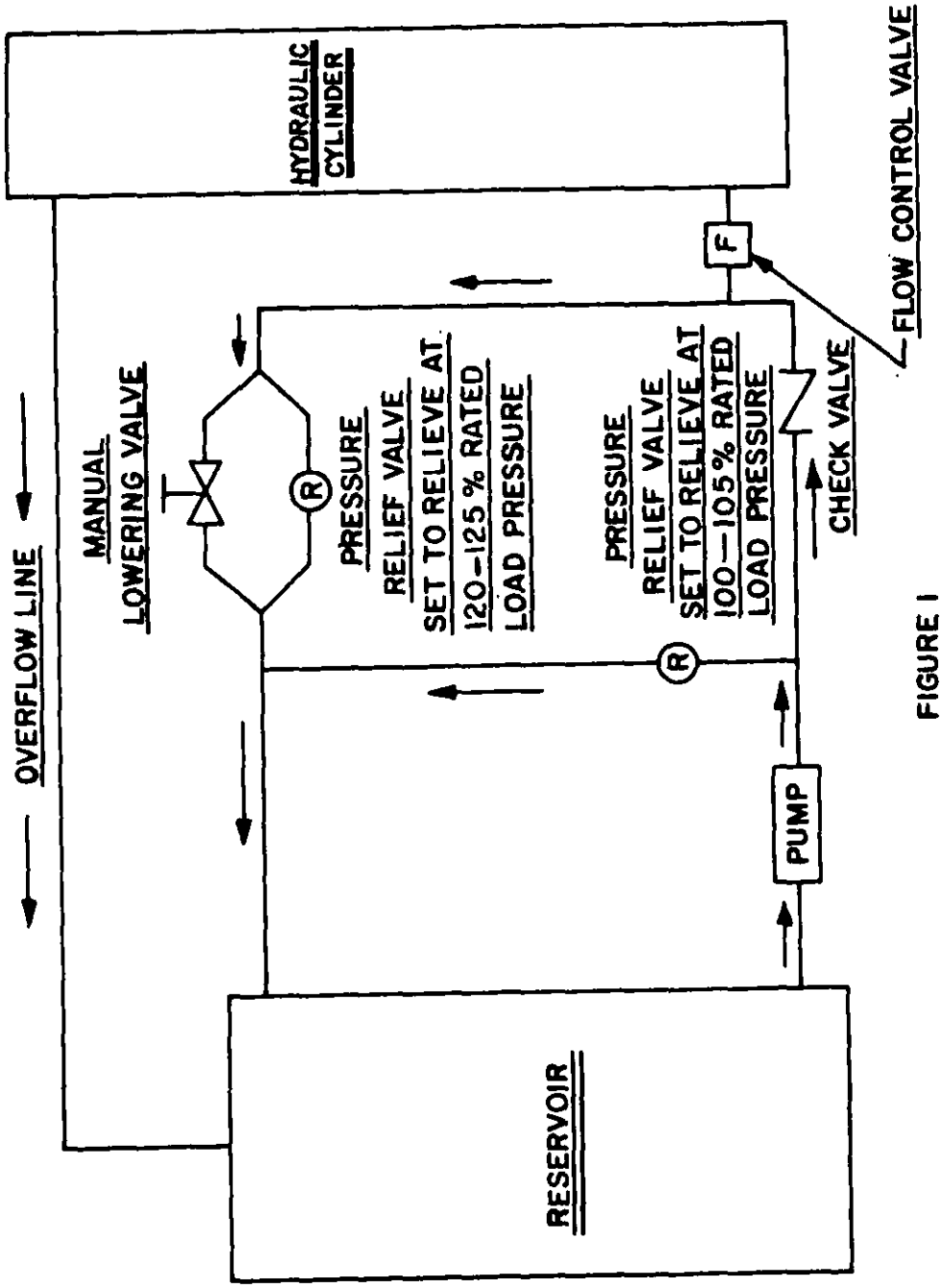


FIGURE 1

TYPICAL HYDRAULIC CRANE CONTROL SYSTEM

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125 percent of capacity when the boom is level and not extended, without permanent deformation of any part of the crane. The safe working capacity of style (a) cranes with boom extended and level in accordance with 3.6.5 shall be not less than 60 percent of the crane's specified working capacity. The crane shall show no evidence of instability or permanent deformation under any conditions of loading set forth above.

* 3.8 Weight. The weight of Group A cranes shall not exceed 850 pounds. The weight of Group B cranes shall not exceed 800 pounds. The weight of Group C cranes shall not exceed 650 pounds.

* 3.9 Treatment and painting. The crane shall be treated and painted in accordance with the manufacturer's standard practice, consistent with good quality, to preclude rust and corrosion.

3.10 Data plate. A fixed plate indicating the safe working capacities of the crane with the boom in various leg positions shall be attached to the crane.

3.11 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130. The nameplate shall conform to MIL-P-514.

3.12 Workmanship. Workmanship shall be of the quality necessary to produce a crane of general good appearance, and possessing the efficiencies of complete and instantaneous operating control characteristics as found in high grade commercial quality items of similar nature.

3.12.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be by methods that will not cause injury to the metal. Shearing and chipping shall be neat and accurate. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best manner. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Precaution shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.12.2 Machine work. Tolerances shall conform to the standards of best commercial practices. Finished contact and bearing surfaces shall be true and exact.

3.12.3 Bolt connections. Boltholes shall be accurately punched and reamed or drilled, and shall have all burrs removed. Bolt threads shall not extend into boltholes when in assembled position. Flat washers shall be used to preclude projection of threaded portion of bolts into bearing surface of boltholes. All bolts, nuts and screws shall be tight.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Components and material inspection. The supplier is responsible for insuring that components and materials are manufactured, examined and tested in accordance with referenced specifications and standards.

* 4.2 Classification of inspection.

a. Preproduction inspection.
b. Quality conformance inspection.

* 4.2.1 Preproduction inspection. Unless otherwise specified (see 6.2), preproduction inspection shall be applied to one sample from each Group, Type, Class and Style of crane that the supplier proposes to furnish the Government under this contract. Failure of the preproduction sample to pass the examination or one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), preproduction inspection shall consist of the examination in 4.3 and all tests in 4.4.

* 4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to cranes prior to being offered for acceptance under the contract. Failure of the crane to pass the examination or one or more tests shall be cause for rejection. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the following:

a. Examination (4.3).
b. Tests (4.4).
c. Inspection of preservation, packaging, packing and marking for shipment and storage. (4.5)

* 4.2.2.1 Sampling plan for quality conformance testing. All cranes of the same group, type, class and style offered for delivery at one time shall be considered a lot for purposes of quality conformance testing. A random sample of cranes shall be selected from each lot in accordance with inspection level S-4 of MIL-STD-105. Each crane contained in the random sample shall be tested as specified in 4.4 and all subparagraphs thereto.

* 4.3 Examination. Each crane shall be examined to determine compliance with requirements specified herein with respect to parts, materials, workmanship, painting, markings and preparation for delivery.

4.4 Tests.

4.4.1 Weight. The crane shall be weighed to verify conformance with 3.8.

4.4.2 Dimensions. The crane shall be measured to verify conformance to all specified dimensions.

4.4.3 Operational test. The crane shall be operated at specified capacities to verify specified capabilities. Adjustments and capabilities of extensible boom for style (a) cranes, and of adjustable legs for class 2 cranes, shall be tested.

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* 4.4.4 Load test. The crane shall be load tested to verify conformance with 3.7. The crane shall show no evidence of permanent deformation of any part and no evidence of instability during the test. Relief valves shall be rendered inoperative during these tests and loads shall be applied for not less than five minutes under each condition.

* 4.4.5 Hoisting test for type I cranes. The hoisting mechanisms of type I cranes shall be examined and tested to verify conformance with 3.6.1.1. The test shall include hoisting a capacity load to maximum height, towing the crane a minimum distance of 30 feet with the load suspended, and lowering the load. The test shall also include the above procedure with a 50 pound load.

* 4.4.6 Hydraulic system tests for type II cranes.

* 4.4.6.1 Leakage test. Relief valves shall first be adjusted as specified in 3.6.2.4. A capacity load shall be lifted and held suspended by the crane for a period of 30 minutes. Downward drift of the load shall not exceed 1/2 inch during this period. There shall be no evidence of external leakage, deformation, or failure of any part of the hydraulic system during this test.

* 4.4.6.2 Hand pump test. The hydraulic hand pump shall be tested to verify conformance with 3.6.2.2.

4.4.6.3 Lowering valve test. The manual lowering valve shall be tested by using it to lower a capacity load. It shall be tested as to its ability to control the rate of lowering from 0 feet per minute through any desired rate up to the maximum lowering speed. The above test shall be repeated with a 50 pound load.

* 4.4.6.4 Flow control valve test. A rated capacity load shall be lifted by the crane and a sudden loss of pressure shall be induced in the system. A sudden dropping of the load shall not result, the speed of lowering being within safe limits.

4.4.6.5 Low pressure relief valve test. In addition to the test specified in 4.4.6.1, an attempt shall be made to lift a load 5 percent greater than rated capacity. The low pressure relief valve shall be open, by-passing the pumped oil back to the reservoir instead of lifting the load.

* 4.4.6.6 High pressure relief valve test. With the boom horizontal, a load 20 percent greater than rated capacity shall be suspended from the boom and held for a period of 30 minutes. Downward drift of the load shall not exceed 1/2 inch during this period. A load increment shall then be added to increase the suspended load to 25 percent greater than rated capacity. The high pressure relief valve shall open, lowering the load.

* 4.5 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2).

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5.1.1.1 Disassembly. Those components, the removal of which is required to reduce the shipping cube to a minimum and which can be reassembled by unskilled personnel using common tools, shall be removed. Bolts, nuts, washers and pins removed shall be replaced in the mating part and secured to prevent loss. Other small parts removed shall have match-marking identification tags attached. These tags shall conform to UU-T-81, type optional.

5.1.1.2 Cleaning and drying. Immediately prior to preservation and packaging, the crane shall be thoroughly cleaned in accordance with Method C-1 and dried in accordance with any drying procedure of MIL-P-116.

5.1.1.3 Preservatives. Preservatives shall conform to MIL-P-116.

5.1.1.3.1 Machined surfaces. Unpainted exterior machined surfaces of component parts shall be coated with type P-2 preservative. The parts involved shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, Type II, Grade A.

5.1.1.3.2 Cables, chains, sprockets, hooks, pins, etc. Unpainted exterior metal surfaces of component parts such as cables, chains, sprockets, hooks, pins, etc., shall be coated with type P-1 preservative. The full length of chain or cable shall be coated.

5.1.1.3.3 Other exposed unpainted surfaces. Coat all other exposed unpainted metal surfaces with type P-1 preservative.

5.1.1.3.4 Hydraulic system. Fill the hydraulic system with hydraulic oil of an operational type recommended by the crane manufacturer. Suitably plug any vent holes and make any other provisions necessary to prevent leakage of the hydraulic oil during shipment or storage.

5.1.1.4 Unit packaging. The crane and component parts shall be unit packaged in accordance with Method I of MIL-P-116.

5.1.2 Level C. Cranes shall be preserved and packaged in accordance with the manufacturer's standard commercial practice. Protection shall be such as to prevent damage or deterioration during shipment to the first receiving activity.

5.2 Packing (see 6.2).

5.2.1 Levels A and B. The crane and component parts shall be packed in a close-fitting nailed wood box conforming to PPP-B-621, Class 2, Style 2, or in a cleated-plywood box conforming to PPP-B-601, overseas type, Style A or B. The box shall be modified by the installation of two by four skids placed across the bottom at right angles to the bottom boards. The contents shall be cushioned, blocked and braced in accordance with MIL-STD-1186. The box shall be closed in accordance with the applicable box specification.

5.2.2 Level C. The crane and component parts shall be packed to insure carrier's acceptance and safe delivery to destination in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2), packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The cranes covered by this specification are intended to be used for straddle-lifting jobs such as removing engines from automotive vehicles.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Group, Type, Class and style of crane required (see 1.2).
c. Grade of lubricants and temperature range required (see 3.3.2 and 3.4.1).
d. Specify whether alloy or steel chain or wire rope is required (see 3.5.5).
e. Quantity of preproduction samples and preproduction inspection, if different (see 4.2.1).
f. Quality conformance inspection, if different (see 4.2.2).
g. Level of preservation and packaging, and packing required (see 5.1 and 5.2).
h. Special marking required (see 5.3).

6.3 Equipment and installation. All miscellaneous equipment and special tools necessary for the proper operation of the crane will be furnished at no additional cost to the purchaser.

* 6.4 Preproduction sample. The preproduction sample may be a preproduction item or a production item which meets the requirement of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a preproduction sample shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 6.5 Notations. The margins of this specification are marked with an asterisk (*) to indicate where changes (additons, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ME
Navy - YD
Air Force - 84

Preparing activity:
Air Force - 84

Reviewers:
Army - GL
DSA - CS

Project number:
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Users:
Navy - SH, SA, MC

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE.

AFLC RAFB,GA JUN 67 - 500

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